

The use of stem cells in treating xerostomia: a systematic review.

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The complex nature of xerostomia prevents the establishment of a definite cure. Recently, research has pivoted towards stem cell transplantation for glandular reconstruction. The aim of this study is to provide an updated review of the existing research, to highlight the encountered challenges and research pathways, potentially enhancing the therapeutic applications of stem cell transplantation. This is a systematic review according to PRISMA guidelines, using the following databases: PubMed (PMC), PMC Europe, Scopus, Medline, Research Gate, Elsevier. The main question was whether stem cell therapies can contribute to the treatment of xerostomia. 19 of 226 publications met the criteria for this review, including 'in vivo', 'in vitro' studies and clinical trials. All 19 studies described thoroughly the stem cell source and the transplantation method, and documented results based on analytical and statistical methods of confirmation. Data show that the various sources of stem cells play a significant role, with bone marrow or adipose tissue-derived pluripotent blasts being the most utilized. Human transplants in mice have also been accepted and reversed hyposalivation. The effects have been beneficial especially in models undergone radiotherapy (IR) or exhibit Sjogren Syndrome-like symptoms (SS), suggesting that with appropriate treatment and enrichment techniques, stem cell transplantation seems effective regardless of the cause of the disorder. Extracts and co-cultures of gland and stem cells also seem to improve gland function. Although in its initial stages, the use of stem cells seems to be a promising therapy to alleviate xerostomia regardless of its cause.

Long-Term Outcome Following Treatment With Allogeneic Mesenchymal Stem/Stromal Cells for Radiation-Induced Hyposalivation and Xerostomia.

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Adipose-derived mesenchymal stem/stromal cells (ASCs) are proposed as a new xerostomia treatment. The study evaluated the long-term safety and effectiveness of allogeneic ASCs in radiation-induced xerostomia among patients with previous oropharyngeal cancer. This study constitutes 3-year follow-up on the original 10 patients who received allogeneic ASCs injections to the submandibular and parotid glands as part of the MESRIX-II trial. The MESRIX-II trial included the preliminary 4-month follow-up. The primary endpoint was long-term safety. Secondary endpoints were effectiveness evaluated by changes in salivary flow rate and patient-reported outcomes (PROs). Immune response was evaluated by assessing the development of donor-specific antibodies (DSA). All 10 MESRIX-II patients completed the long-term follow-up (ie, no missing data). During the long-term follow-up, 2 patients encountered a significant adverse event, which was determined to be unrelated to the treatment. No DSAs were detectable at 3 years. The stimulated salivary flow rate increased significantly from an average of 0.66 mL/minute at baseline to 0.86 mL/minute at follow-up, corresponding to an increase of 0.20 [95% CI 0.08 to 0.30] mL/minute, or approximately 30%. Among the PROs, sticky saliva symptoms were reduced, with a -20.0 [95% CI -37.3 to -2.7] units. In conclusion, this study is the first to present long-term follow-up outcomes of allogeneic ASC treatment as a therapeutic option for radiation-induced xerostomia. The study found that ASC treatment appears safe, and there were no indications of adverse immune responses at the 3-year follow-up. Further studies are warranted to evaluate the findings in larger settings.

Efficacy of mesenchymal stem cell therapy in rodent models of radiation-induced xerostomia and oral mucositis: a systematic review.

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Radiation-induced xerostomia and oral mucositis are serious complications of radiation therapy for head and neck cancers. Current treatment options have limited efficacy. Mesenchymal stem cell (MSC) therapy has shown promising results in supporting the restoration of glandular secretion function and the regeneration of damaged tissues. This study aim to (1) assess the quality of evidence for MSCs treatment in rodent models of radiation-induced oral complications and (2) determine whether MSCs can improve the therapeutic effect of radiation-induced oral mucositis. Intervention studies using MSCs in rodent models were comprehensively retrieved in the Pubmed, Medline, Embase, Web of Science, and Cochrane library databases on June 1, 2022. The quality of all in vivo experiments was assessed using SYRCLE, and this article is written following the PRISMA guidelines. A total of 12 studies were included in this systematic review. The study found that in animal models of radiation-induced xerostomia, MSCs could increase salivary protein secretion, improve SFR, shorten the salivary lag time, anti-apoptosis, etc. In animal models of radiation-induced oral mucositis, MSCs improve the micromorphology and macromorphology of RIOM. Moreover, the effect of MSCs on the modification of ulcer duration and latency may be related to the time of MSCs transplantation but further studies are needed. The results of our systematic review suggest that MSCs appeared to be effective in the treatment of radiation-induced xerostomia and oral mucositis.

Long-term Safety of Treatment with Autologous Mesenchymal Stem Cells in Patients with Radiation-Induced Xerostomia: Primary Results of the MESRIX Phase I/II Randomized Trial.

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Mesenchymal stem/stromal cell therapy may reduce radiation-induced xerostomia. We investigated the long-term safety of autologous adipose tissue-derived mesenchymal stem/stromal cell (ASC) injections into the submandibular glands. An investigator-initiated, randomized, single-center, placebo-controlled trial. Previous patients with oropharyngeal squamous cell carcinoma with radiation-induced xerostomia were randomly (1:1) allocated to receive a 2.8 million ASCs/cm³ injection or placebo in both submandibular glands and followed for a minimum of 2 years. The primary endpoint was number of serious adverse events (SAE). Secondary endpoints included whole saliva flow rates and xerostomia-related symptoms. Data analysis was based on the intention-to-treat population using repeated measures mixed-effects linear models. Thirty-three patients were randomized; 30 patients were treated (ASC group, n = 15; placebo group, n = 15). Long-term safety data were collected from all 30 patients. During follow-up, 6 of 15 (40%) of the ASC-treated patients versus 5 of 15 (33%) of the placebo patients experienced an SAE; no SAEs appeared to be treatment related. Unstimulated whole saliva flow rate increased to 0.20 and 0.16 mL/minute in the ASC and placebo group, respectively, yielding a 0.05 mL/minute (95% confidence interval: 0.00-0.10; P = 0.051) difference between groups. Patient-reported xerostomia symptoms diminished according to a decreased xerostomia questionnaire summary score of 35.0 and 45.1, respectively [-10.1 (-18.1 to -2.2); P = 0.013]. Three of the visual analog scale xerostomia measures indicated clinical benefit following use of ASC. Our data show that ASC therapy is safe with a clinically relevant effect on xerostomia-related symptoms. Confirmation in larger randomized controlled trials is warranted.

Safety and Efficacy of Mesenchymal Stem Cells for Radiation-Induced Xerostomia: A Randomized, Placebo-Controlled Phase 1/2 Trial (MESRIX).

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Salivary gland hypofunction and xerostomia are major complications to head and neck radiotherapy. This trial assessed the safety and efficacy of adipose tissue-derived mesenchymal stem cell (ASC) therapy for radiation-induced xerostomia. This randomized, placebo-controlled phase 1/2 trial included 30 patients, randomized in a 1:1 ratio to receive ultrasound-guided transplantation of ASCs or placebo to the submandibular glands. Patients had previously received radiotherapy for a T1-2, N0-2A, human papillomavirus-positive, oropharyngeal squamous cell carcinoma. The primary outcome was the change in unstimulated whole salivary flow rate, measured before and after the intervention. All assessments were performed one month prior (baseline) and one and four months following ASC or placebo administration. No adverse events were detected. Unstimulated whole salivary flow rates significantly increased in the ASC-arm at one (33%; $P = .048$) and four months (50%; $P = .003$), but not in the placebo-arm ($P = .6$ and $P = .8$), compared to baseline. The ASC-arm symptom scores significantly decreased on the xerostomia and VAS questionnaires, in the domains of thirst (-22%, $P = .035$) and difficulties in eating solid foods (-2%, $P = .008$) after four months compared to baseline. The ASC-arm showed significantly improved salivary gland functions of inorganic element secretion and absorption, at baseline and four months, compared to the placebo-arm. Core-needle biopsies showed increases in serous gland tissue and decreases in adipose and connective tissues in the ASC-arm compared to the placebo-arm ($P = .04$ and $P = .02$, respectively). MRIs showed no significant differences between groups in gland size or intensity ($P < .05$). ASC therapy for radiation-induced hypofunction and xerostomia was safe and significantly improved salivary gland functions and patient-reported outcomes. These results should encourage further exploratory and confirmatory trials.

Effectiveness and safety of mesenchymal stem/stromal cell for radiation-induced hyposalivation and xerostomia in previous head and neck cancer patients (MESRIX-III): a study protocol for a single-centre, double-blinded, randomised, placebo-controlled, phase II study.

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A predominant side effect of radiotherapy for head and neck cancer is salivary gland hypofunction and xerostomia leading to debilitating oral disorders and impaired quality of life (QoL). Intraglandular mesenchymal stem cell therapy has shown promising results as a treatment for xerostomia. This is a randomised, double-blinded, placebo-controlled, parallel-group, prospective, single-centre trial investigating the safety, tolerability, and effectiveness of allogeneic stem cells as a treatment for radiation-induced hyposalivation and xerostomia for previous head and neck cancer patients. We will include a total of 120 patients who previously have been treated with radiotherapy for a head and neck cancer in Denmark. Participants will be randomly assigned using block randomisation to one of two parallel groups in a 1:1 ratio to receive ultrasound-guided injection of allogeneic adipose-derived mesenchymal stem cell (ASC) ($n = 60$) or placebo ($n = 60$) into the submandibular glands. Placebo will consist of CryoStor10 (BiolifeSolutions), the freeze media for ASCs containing 10% dimethyl sulfoxide (DMSO). The primary endpoint is change in unstimulated whole saliva flow rate. The secondary endpoints

are change in stimulated whole saliva flow rate, QoL, and composition of saliva. Further secondary endpoints are safety and immune response (human leukocyte antigen (HLA) response) to the stem cells will be assessed. Patients are evaluated at baseline (before treatment), after 4 months, and after 12 months. All study personnel, except study personnel thawing and preparing the treatment for injection, and participants will be blinded to group assignment. Unblinded study personnel will not participate in the outcome assessment. The trials will investigate the efficacy and safety of ASC injection to the submandibular gland as a potential new treatment for post-radiation xerostomia. We hope the results will pave the way for a clinically relevant treatment to ameliorate patients with xerostomia, a severely hampering condition. The study is approved by the Danish Data Protection Agency (protocol number P-2020-1164), the National Ethics Committee protocol number: (Protocol number: 1802872), and the Danish Medical Agency (2018-000348-24). The protocol was registered at the ClinicalTrials.gov database (NCT04776538).

Marrow-Derived Autologous Stromal Cells for the Restoration of Salivary Hypofunction (MARSH): Study protocol for a phase 1 dose-escalation trial of patients with xerostomia after radiation therapy for head and neck cancer: MARSH: Marrow-Derived Autologous Stromal Cells for the Restoration of Salivary Hypofunction.

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Xerostomia, or dry mouth, is a common side effect of head and neck radiation. Current treatment options for radiation-induced xerostomia are generally supportive in nature. Adult stem cells are the ultimate source for replenishment of salivary gland tissue. Bone marrow-derived mesenchymal stromal cells (BM-MSCs) are a viable cell-based therapy for xerostomia. We have undertaken studies enabling U.S. Food and Drug Administration Investigational New Drug status, demonstrating the normal phenotype, intact functionality, and pro-growth secretome of interferon- γ (IFN γ)-stimulated BM-MSCs taken from patients with head and neck cancer who have undergone radiation $\pm$ chemotherapy. Here we present the protocol of MARSH, a first-in-human clinical trial of bone marrow-derived, IFN γ -activated BM-MSCs for the treatment of radiation-induced xerostomia. This single-center phase 1 dose-escalation with expansion cohort, non-placebo-controlled study will assess the safety and tolerability of BM-MSCs for the treatment of radiation-induced xerostomia in patients who had head and neck cancer. The phase 1 dose-escalation study will be a 3 + 3 design with staggered enrollment. A total of 21 to 30 subjects (9 to 18 in phase 1 study, 12 in expansion cohort) will be enrolled. The primary endpoint is determining the recommended phase 2 dose (RP2D) of IFN γ -stimulated BM-MSCs to enable further studies on the efficacy of BM-MSCs. Patients' bone marrow will be aspirated, and BM-MSCs will be expanded, stimulated with IFN γ , and injected into the submandibular gland. The RP2D will be determined by dose-limiting toxicities occurring within 1 month of BM-MSC injection. Secondary outcomes of saliva amounts and composition, ultrasound of salivary glands, and quality of life surveys will be taken at 3-, 6-, 12-, and 24-month visits. Autotransplantation of IFN γ -stimulated BM-MSCs in salivary glands after radiation therapy or chemoradiation therapy may provide an innovative remedy to treat xerostomia and restore quality of life. This is the first therapy for radiation-induced xerostomia that may be curative. World Health Organization International Clinical Trials Registry Platform: NCT04489732.

Post-radiation xerostomia therapy with allogeneic mesenchymal stromal stem cells in

patients with head and neck cancer: study protocol for phase I clinical trial.

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Xerostomia is a common side effect of radiotherapy in patients with head and neck tumors that negatively affects quality of life. There is no known effective standard treatment for xerostomia. Here, we present the study protocol used to evaluate the safety and preliminary efficacy of allogeneic mesenchymal stromal stem cells (MSCs) derived from umbilical cord tissue. Ten oropharyngeal cancer patients with post-radiation xerostomia and no evidence of disease recurrence 2 or more years after (chemo)irradiation (intervention group) and 10 healthy volunteers (control group) will be enrolled in this nonrandomized, open-label, phase I exploratory study. MSCs from umbilical cord tissue will be inserted under ultrasound guidance into both parotid glands and both submandibular glands of the patients. Toxicity of the procedure will be assessed according to CTCAE v5.0 criteria at days 0, 1, 5, 28, and 120. Efficacy will be assessed by measuring salivary flow and analyzing its composition, scintigraphic evaluation of MSC grafting, retention, and migration, and questionnaires measuring subjective xerostomia and quality of life. In addition, the radiological, functional, and morphological characteristics of the salivary tissue will be assessed before, at 4 weeks, and at 4 months after the procedure. In the control group subjects, only salivary flow rate and salivary composition will be determined. The use of allogeneic MSCs from umbilical cord tissue represents an innovative approach for the treatment of xerostomia after radiation. Due to the noninvasive collection procedure, flexibility of cryobanking, and biological advantages, xerostomia therapy using allogeneic MSCs from umbilical cord tissue may have an advantage over other similar therapies.

The role of stem cells in the prevention and treatment of radiation-induced xerostomia in patients with head and neck cancer.

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Xerostomia is an important complication following radiotherapy (RT) for head and neck cancer. Current treatment approaches are insufficient and can only temporarily relieve symptoms. New insights into the physiopathology of radiation-induced xerostomia might help us in this regard. This review discusses the current knowledge of salivary gland stem cells in radiation-induced xerostomia and their value in the prevention and treatment of this complication. Salivary gland stem cell transplantation, bone marrow-derived cell mobilization, molecular regulation of parotid stem cells, stem cell sparing RT, and adaptive RT are promising techniques that are discussed in this study.

Clinical Introduction of Stem Cell Sparing Radiotherapy to Reduce the Risk of Xerostomia in Patients with Head and Neck Cancer.

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Studies have shown that dose to the parotid gland stem cell rich (SCR) regions should be reduced to lower the risk of xerostomia after radiotherapy (RT). This study aimed to assess whether stem cell sparing (SCS)-RT can be adopted in routine clinical practice. Multiple planning studies were performed to compare SCS-RT with standard (ST)-RT using 30 head and neck cancer patients. Shifts in mean dose to the SCR regions (Dmean,SCR) and other organs at risk and their estimated impact on normal tissue

complication probability (NTCP) for side-effects were compared using Wilcoxon signed-rank test. A multicenter study was performed (eight institutions, three patients) to test the generalizability of SCS-RT using the Friedman test. Using photons, $D_{mean,SCR}$ was reduced with median 4.1/3.5 Gy for ipsilateral/contralateral ($p < 0.001$). The largest reductions were when the SCR regions overlapped less with target volumes. Subsequently, NTCPs for xerostomia decreased ($p < 0.001$). Using protons, $D_{mean,SCR}$ was also reduced (2.2/1.9 Gy for ipsilateral/contralateral, $p < 0.002$). Nevertheless, SCS-RT did not further decrease NTCPs for xerostomia ($p > 0.17$). Target coverage and prevention of other side-effects were not compromised. However, increased mean oral cavity dose was observed in some patients. Lastly, in the multicenter study $D_{mean,SCR}$ could be reduced by slightly adjusting the standard optimization. Contralateral $D_{mean,SCR}$ reductions differed between centers ($p = 0.01$), which was attributed to differences in ST-RT plans. Stem cell sparing radiotherapy can be clinically introduced by making small adjustments to the optimization strategy and can reduce the risk of xerostomia.
